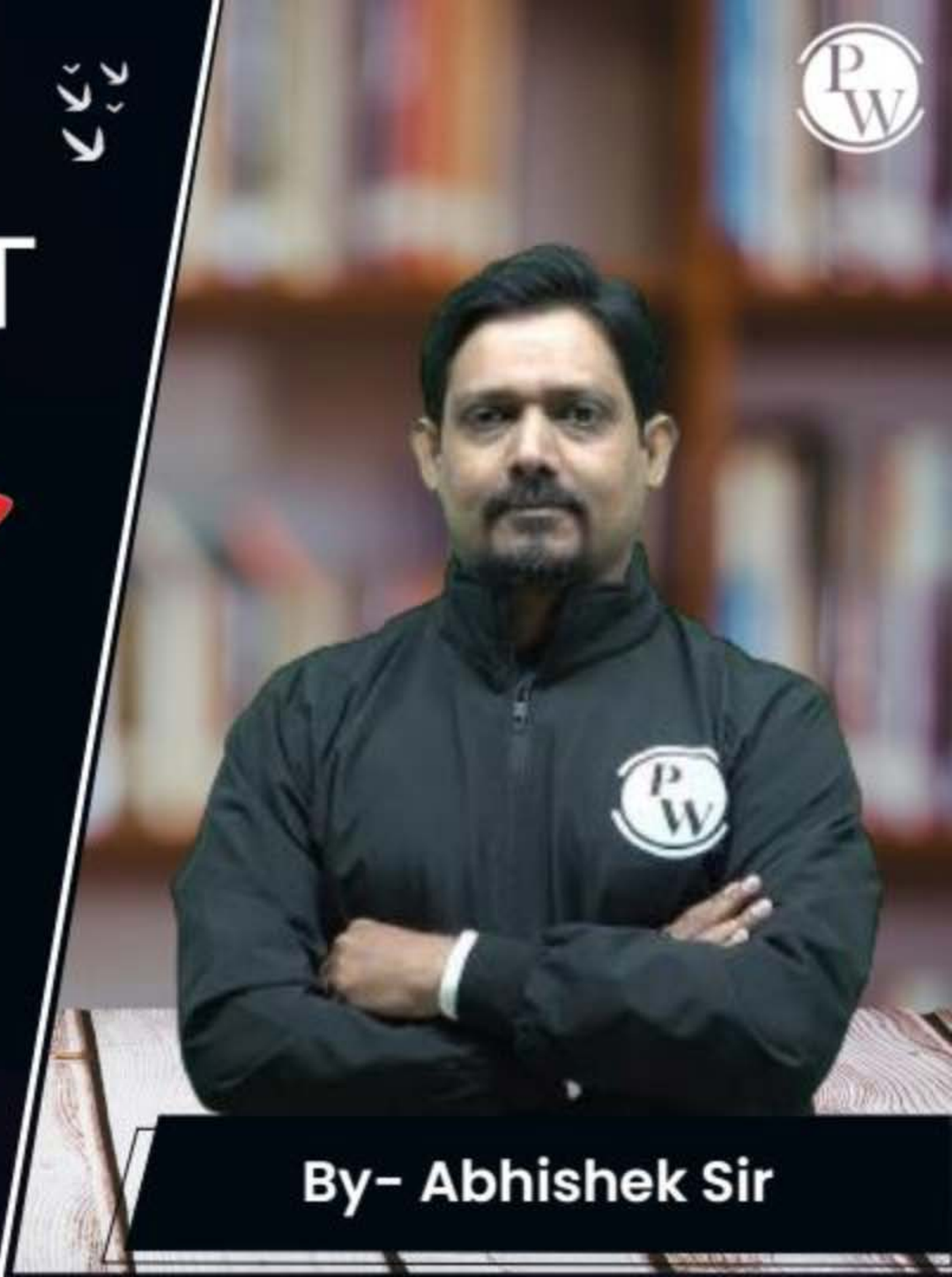
Computer Science & IT



Fundamental of C Language

Array & Pointers

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



Topic — Array

Topic — Array declaration

Topic — Array intalization

Topic — Array element Access

Topic

Slide

Topics to be Covered



- Topic — Address calculation of 1-D array
- Topic — 2-D array
- Topic
- Topic
- Topic

Slide



Array Addressing

Memory used to store array element

Sequential manner :- int a[10] = {1,2,3,4,5,6,7,8,9,10};

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]
1	2	3	4	5	6	7	8	9	10

Slide



Array Addressing

A[0] is first element

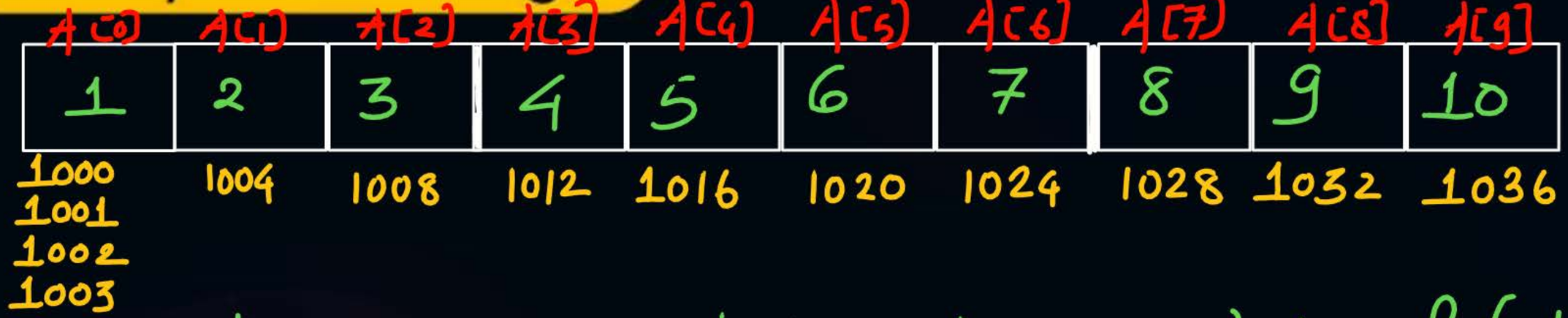
first element starts with index 0

Each Memory location is having an address

Address: Byte Addressable ∴ 1 Byte is assigned 1 Address

Slide

Array Addressing



Base Address: The starting Address or Address of first element of array is Base Address.

Lets assume array starts from 1000

Size of each element : Because its an integer array Size = 4 Bytes

Slide



Array Addressing

int a[50] = {1,2,3,4.....50};

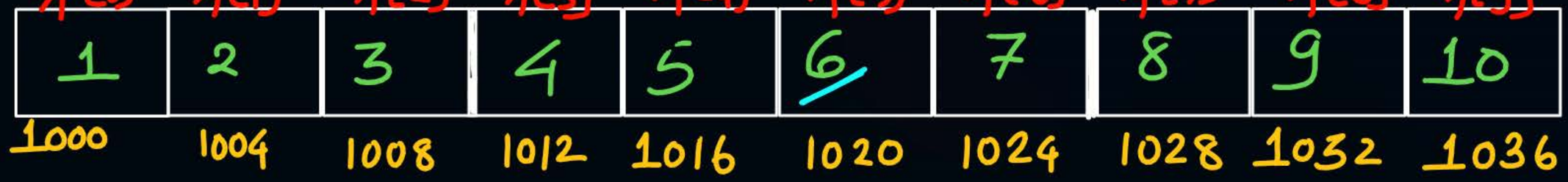
Question : BA is 1000 , Size = 4B then Address of A[30]

Answer :- BA + i x S

= 1000 + 30 x 4 : 1000 + 120 = 1120

Slide

Array Addressing



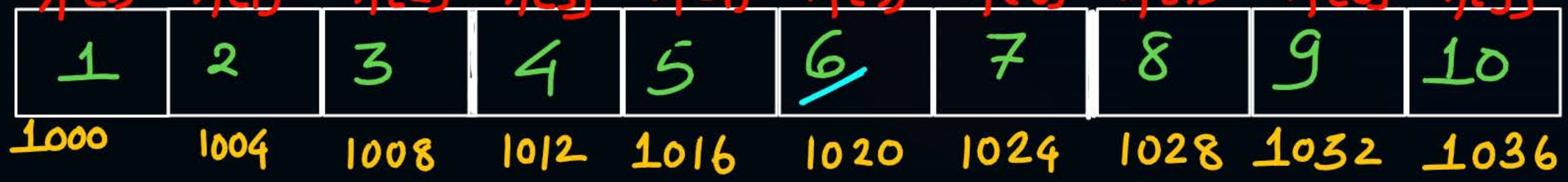
Suppose we want to find Address of A[5]

Before A[5], No. of element arranged : 5

Address of A[5] = 1000 + 5×4 = 1020

Slide

Array Addressing



Suppose we want to find Address of A[9]

Before A[9] No. of element arranged = 9

1000 + 9 × 4 = 1036

Slide



Array Addressing

Suppose BA is Base Address & S is Size

Calculate Address of A[i]

Before A[i], No. of elements arranged i

$$\text{Address of } A[i] = BA + i \times \text{Size}$$



Array Addressing

Array of character :- `char ch[4] = {'a','b','c','d'};`

`char ch1 = 'a';`

Slide



Array Addressing

Q: Suppose A character Array A[100] is declared if BA is 230, Size is 1Byte. What is the Address of A[73]?

Ans

Address of A[i] = BA + i x S

BA is 230

Size : 1Byte

i = 73

= 230 + 73 x 1 = 303

Slide



Array Addressing

double d[10] = {1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 7.7, 8.8, 9.9, 10.11};

double - 8 Byte

Question :- Suppose An array of double is declared as double d[10]

BA is 540 Size is 8 Byte What is the Address of A[79]?

Ans. 1172

Slide



Array Addressing

Address of A[i] = BA + i x S

BA = 540

S = 8 Byte

i = 79

A[i] = 540 + 79 x 8

= 540 + 632

= 1172 ✓

Slide



2-D Array

Dimension

```
int a[4][5];
```

Declaration of 2-D array



2 mins Summary

Topic 1-D array

Topic Memory arrangment

Topic Byte Addressable

Topic Address of A[i]

Topic

Slide

THANK - YOU

